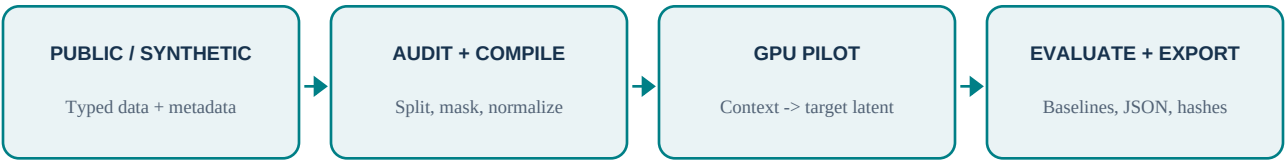


JEPA-FORGE

Prototype implementation and GPU experiment report

A reproducible feasibility pilot for scientific dataset-to-task compilation

This working prototype implements a typed task compiler, train-only normalization, structural/leakage diagnostics, numeric exports with checksums, a context-only inference API, and small JEPA-style GPU pilots. It does not implement the complete proposed ecosystem or reproduce the proposal's earlier result table.



Train-only preprocessing | disjoint views | held-out groups | context-only inference

What was executed

18 neural runs cover three datasets and six policies, with model seeds 7, 17 and 27, a fixed split/generator seed of 2026, and 60 epochs per run. The first declared policy for each dataset was designated primary before evaluating the test set. Sample standard deviations measure model-seed variation on one fixed split, not population uncertainty.

Data	Task family	Evaluation
Wine: 178 rows, 13 features	Feature-group prediction	Three-class frozen linear probe
Digits: 1,797 images, 8 x 8 pixels	Spatially masked prediction	Ten-class frozen linear probe
Synthetic grouped sensor windows	Past-to-future prediction	Future-signal regression

Measured primary results

wine / alternating: JEPA macro_f1 0.914 +/- 0.000; raw-context linear 0.941 +/- 0.000.

digits / center: JEPA macro_f1 0.886 +/- 0.008; raw-context linear 0.875 +/- 0.000.

synthetic_sensors / past16_future8: JEPA rmse 0.321 +/- 0.021; raw-context linear 0.259 +/- 0.000.

Evidence files: results/benchmark.json, results/environment.json, results/validation.json. JSON manifests record split counts and hashes; exact split indices reside in local dataset.npz exports.

Data integrity and model design

Compiler contract

RawDataset retains feature names, numeric observations, modality/shape, labels, group identities, interval coverage and provenance. TaskSpec records disjoint observed and hidden indices. Labels and identifiers are prohibited from task inputs when declared in the schema.

Check	Purpose / boundary
Schema and finite numeric data	Reject malformed shapes, invalid feature indices and non-finite observations.
Split and group integrity	Reject reused row indices, repeated source groups and duplicate records across splits.
Context-target structure	Reject overlap and reverse-time prediction; retain spatial/sequence layouts.
Train-only normalization	Fit mean and scale on training rows; reuse those parameters for validation and test.
Export hashes	Verify JSON and numeric NPZ bytes; no pickle loading. Integrity does not authenticate provenance.

JEPA-style training

The online encoder processes observed values plus a binary mask. A separate frozen teacher processes the hidden target view. An MLP predictor maps context embeddings to target embeddings. The teacher uses an exponential moving average of the online encoder. Training combines latent MSE with a context-variance penalty; the latter is an explicitly disclosed prototype adaptation.

An MLP handles Wine, a small Conv2d encoder handles Digits, and a Conv1d encoder preserves sensor order. Inference accepts only standardized context columns. The public context encoder API cannot consume full rows or concatenate hidden-target embeddings.

Evaluation boundaries

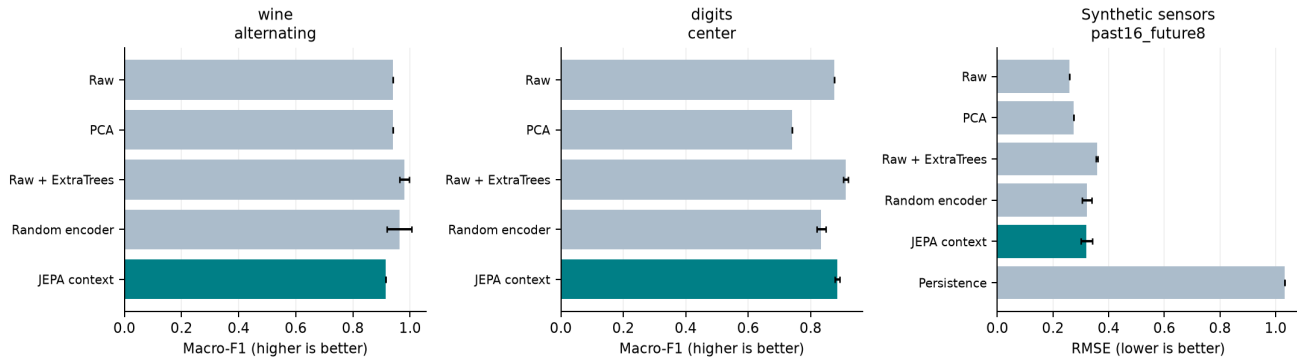
All methods share saved train/validation/test indices. Linear-probe hyperparameters and ExtraTrees depth are selected on validation data. PCA and standardization fit training data only. Model training uses the predeclared final epoch. The test partition is not used to train, normalize, select checkpoints or choose task policies.

Targets have different information budgets across policies: central masking hides 16 pixels, right-half masking hides 32, and sensor forecasting uses 8- or 12-step horizons. Compare methods within one policy. No cross-policy performance ranking is asserted.

Primary task results

Context-only predictions; mean +/- sample SD across seeds 7, 17 and 27. Classification uses macro-F1 (higher is better). Forecasting uses RMSE in original synthetic units (lower is better).

Context-only evaluation: identical visible inputs within each task



Method	Wine / F1	Digits / F1	Sensors / RMSE
Raw context + linear	0.941 +/- 0.000	0.875 +/- 0.000	0.259 +/- 0.000
PCA + linear	0.941 +/- 0.000	0.741 +/- 0.000	0.274 +/- 0.000
Raw context + ExtraTrees	0.980 +/- 0.018	0.913 +/- 0.008	0.358 +/- 0.004
Random encoder + linear	0.962 +/- 0.044	0.834 +/- 0.014	0.322 +/- 0.018
JEPA context + linear	0.914 +/- 0.000	0.886 +/- 0.008	0.321 +/- 0.021
Persistence	n/a	n/a	1.033 +/- 0.000
Full raw reference*	0.970 +/- 0.000	0.947 +/- 0.000	n/a

* Full raw reference sees every original feature or pixel, including the hidden region. It is reported separately as a full-information reference; it is not a matched comparator. n/a means the method does not apply.

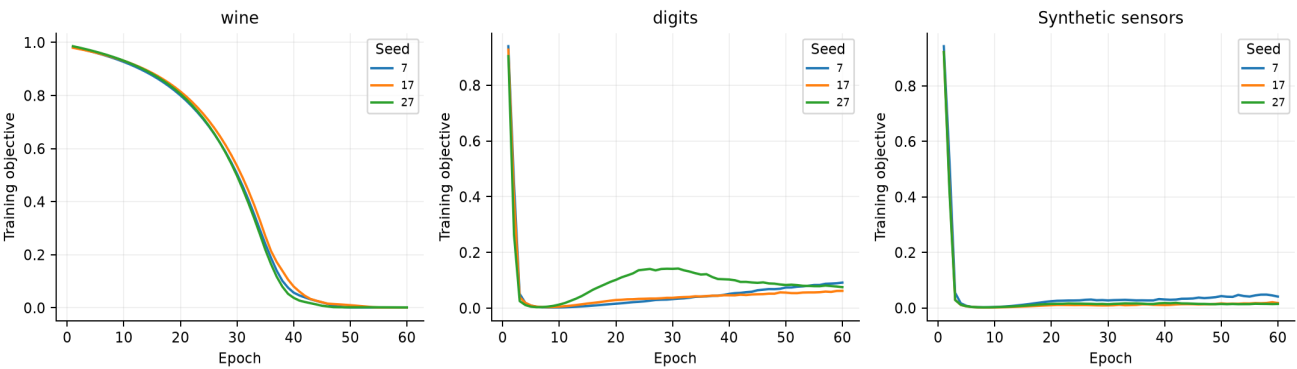
Interpretation

On the primary tasks, JEPA improved Digits macro-F1 over the raw-context linear probe, but remained below ExtraTrees. On Wine, JEPA underperformed both raw-context baselines. For sensors, JEPA beat persistence but had higher RMSE than raw-context ridge. The results support pipeline feasibility and expose model limitations; they do not establish general JEPA superiority.

The same dataset split is used for all three model seeds. Deterministic linear baselines can have zero seed standard deviation. These are repeated optimization runs, not independent replications of the data-generating population.

Alternative tasks and training evidence

Alternative policy	Metric	JEPA	Raw linear	ExtraTrees
wine / first_half	macro_f1	0.907 +/- 0.032	0.892 +/- 0.000	0.990 +/- 0.018
digits / right_half	macro_f1	0.870 +/- 0.004	0.848 +/- 0.000	0.921 +/- 0.003
synthetic_sensors / past12_future12	rmse	0.404 +/- 0.042	0.339 +/- 0.000	0.451 +/- 0.001



Compute evidence

Environment: Apple M3 Max, 40 GPU cores, 128 GB unified memory; PyTorch 2.14.0. The full benchmark elapsed time, including CPU evaluation and exports, was 0.79 minutes. The isolated forward/backward GPU smoke check passed. Every neural run records actual parameter/batch devices and synchronized training timing.

Compiler and test evidence

```
{
  "status": "passed",
  "tests_passed": 44,
  "test_failures": 0,
  "completed_gpu_runs": 18,
  "audited_tasks": 6,
  "export_reload_checks": 6,
  "checkpoint_reload_checks": 18,
  "actual_device": "mps:0",
  "runtime_source_hashes_match": true,
  "low_rank_warning_runs": 7
}
```

Per-seed loss histories, validation predictability diagnostics, effective ranks and embedding standard deviations are preserved in results/benchmark.json. Seven of 18 runs triggered the low-rank heuristic (effective rank below 2): all six Wine runs and one 12-step sensor run. These flags identify a limitation of this pilot model.

Reproduction and limits

Run the prototype

Create a Python 3.11+ environment and install `pip install '[dev,report]'`. Run `pytest -q`, then inspect a dataset with `jepa-forge inspect wine`. Use the listed policy name with `jepa-forge compile`.

Run `python scripts/run_experiments.py --config configs/benchmark.json`, then `python scripts/build_report.py`. The supplied config requires MPS. Explicitly set CUDA for an NVIDIA GPU or CPU for a disclosed CPU run. GPU unavailability fails clearly.

Evidence and portability

The repository includes source, tests, dataset attribution, the experiment protocol, fixed configuration, dependency lock snapshot, measured JSON, charts and this report. Local run artifacts also include compiled NPZ datasets and tensor-only checkpoints. JSON manifests record dataset/split hashes; NPZ archives contain split indices and compiler means/scales. Probe/PCA estimators are reproduced by refitting the recorded grids and seeds. GPU timing includes software/device effects and is not a cross-hardware speed comparison.

What remains to establish

This is a bounded research prototype. Graphs, audio, event streams, multimodal fusion, generalized dataset discovery, untrusted plugin execution, signed releases, ecosystem governance and user studies are not implemented. Leakage rules depend on correct metadata and cannot discover every semantic shortcut.

Wine has a small test set. Digits does not establish writer-disjoint generalization. Synthetic sensor dynamics do not demonstrate real-world industrial forecasting. Additional data splits, datasets, ablations, external replications and domain review are required before broader claims.

Sources and attribution

I-JEPA: arxiv.org/abs/2301.08243. Official implementation: github.com/facebookresearch/ijepa.

PyTorch MPS: docs.pytorch.org/docs/stable/notes/mps.html.

UCI Wine: archive.ics.uci.edu/dataset/109/wine. UCI Optical Digits: archive.ics.uci.edu/dataset/80/optical+recognition+of+handwritten+digits.

Dataset records identify CC BY 4.0; full attribution and loader/subset details appear in `docs/DATASETS.md`.

scikit-learn leakage guidance: scikit-learn.org/stable/common_pitfalls.html.